

ORIGINAL ARTICLE

Dominance of *Methanomicrobium* phylotype in methanogen population present in Murrah buffaloes (*Bubalus bubalis*)

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Abstract

Aims: To study the diversity of rumen methanogens in Murrah buffaloes (*Bubalus bubalis*) from North India by using 16S rRNA gene libraries obtained from the pooled rumen content from four animals and using suitable software analysis.

Methods and Results: Genomic DNA was isolated and PCR was set up by using specific primers. Amplified product was cloned into a suitable vector and the positive clones were selected on the basis of blue–white screening and sequenced. The resulting nucleotide sequences were arranged in the phylogenetic tree. A total of 108 clones were examined, revealing 17 different 16S rRNA gene sequences or phylotypes. Of the 17 phylotypes, 15 (102 of 108 clones) belonged to the genus *Methanomicrobium*, indicating that the genus *Methanomicrobium* is the most dominant component of methanogen populations in Murrah buffaloes (*Bubalus bubalis*) from North India. The largest group of clones (102 clones) was more than 98% similar to *Methanomicrobium mobile*. BLAST analysis of the rumen contents from individual animals also revealed 17 different phylotypes with a range of 3–10 phylotypes per animal.

Conclusion: *Methanomicrobium* phylotype is the most dominant phylotype of methanogens present in Murrah buffaloes (*Bubalus bubalis*).

Significance and Impact of the Study: Effective strategies can be made to inhibit the growth of *Methanomicrobium* phylotype to reduce the methane emission from rumen contents and thus help in preventing global warming.

Introduction

In India, enteric methane is produced by methanogenic archaea as a by-product of fermentation of feed in the gut. One way of reducing global methane emissions is by reducing the number and/or activity of these methanogens. In order to develop successful strategies to reduce enteric methane, we should have information about the composition and distribution of methanogen species in ruminant livestock. In earlier studies (which comprise data from across three continents), *Methanobrevibacter* strains accounted for the majority of methanogens in the rumen of sheep and cattle in Australia, Canada, Japan, New Zealand and the USA (Miller and Wolin 1986; Sharp *et al.* 1998; Whitford *et al.* 2001; Wright and Pimm 2003; Irbis and Ushida 2004; Skillman *et al.* 2004, 2006; Wright *et al.*

2004, 2006). However, to our knowledge, there is no published information available on methanogens from domesticated livestock in India. Therefore, the aim of this study was to use 16S rRNA gene library analysis and some software analysis to investigate the composition of methanogen species from Murrah buffaloes (*Bubalus bubalis*) in India. We hypothesized that *Methanomicrobium* phylotypes are the dominant methanogens in the Murrah buffaloes (*Bubalus bubalis*) from India.

Materials and methods

Animals and diet

Four Murrah buffaloes (*Bubalus bubalis*) of both sexes, approx. 12 months old with an average weight of 700 kg,

were randomly selected. They were individually accommodated in animal house rooms located at National Dairy Research Institute in Karnal, Haryana state, India. Each animal house was provided with a feed bucket and drinking nozzle with adequate amount of water at all times. The animals were individually fed once a day, for 8 weeks before collection of rumen contents. The buffaloes were fed on a standard diet (concentrate/roughage ratio = 40 : 60). Rumen liquor samples from the buffaloes were obtained prior to the first morning feeding.

Source of samples and processing

Rumen liquor sample (approx. 30 ml) was obtained from each animal through a stomach tube attached to a pump, 1 h before morning feeding (0700 h). The samples were squeezed immediately through two layers of muslin cloth to remove any impurities (plant extracts, undigested matter, etc.). The rumen contents were immediately processed for DNA extraction.

DNA extraction, PCR amplification and clone library construction

One millilitre of clarified rumen fluid was subjected to DNA isolation using Bacterial genomic DNA isolation kit (Chromas Biotech Pvt. Ltd., Bangalore, India). All the DNA extractions were carried out on fresh rumen fluid, and the resulting DNA was stored at -20°C until use for PCR. PCR was set up by using Met86F and Met1340R set of primers. The steps in PCR include 95°C for 5 min, 40 cycles of 94°C for 1 min, 58°C for 30 s, 72°C for 1.5 min and a final extension step of 10 min at 72°C . The resulting 1.1 kb PCR product was purified and cloned into a suitable vector by using TOPO-TA cloning kit (Invitrogen). Positive clones were selected on the basis of blue-white screening. Positive clones were propagated and subjected for plasmid isolation (GenElute plasmid miniprep kit (Sigma)). The plasmid DNA was re-amplified using PCR with the same conditions as used previously. The resulting plasmid PCR products were run on 1.5% agarose gel for 1 h at 100 volts, and the samples having the inserts were sent for sequencing (GeneOmniotechnologies, Pune, India). Clones were designated as *Bubalus bubalis* followed by a clone number.

BLAST and CLUSTAL W analyses

Sequences obtained after sequencing were aligned by using CLUSTAL W. The resulting sequences were compared with the Basic Local Alignment Search Tool (BLAST) in order to find out the sequence homology with the methanogen sequences available in the GenBank database.

Phylogenetic analysis

Seventeen additional 16S rRNA gene sequences representing all major methanogen orders were included in the phylogenetic analyses. All sequences were globally aligned using the MEGA 4 software (Tamura *et al.* 2007). *Methanopyrus kandleri* (AE010349) was used as outgroup. MEGA4 11 was used to generate the phylogenetic tree as well as to calculate the bootstrap values.

Nucleotide sequence accession numbers

The sequences from this study have been deposited in the GenBank database under the accession numbers EU360960, EU487513, EU487514, EU487515, EU487516, EU487517, EU487518, EU487519, EU487520, EU487521, EU487523, EU487522 and EU625235.

Reference sequences used in the study

Following are the references used in this study: *Methanomicrobium mobile* (M59142), *Methanocorpusculum bavaricum* (AY196676), *Methanococcoides burtonii* (X65537), *Methanosarcina barkeri* (M59144), *Methanosarcina mazei* (U20151), *Thermoplasma acidophilum* (M38637), *Methanococcus jannaschii* (M59126), *Methanococcus vannielii* (AY196675), *Sulfolobus acidocaldarius* (U05018), *Thermoplasma aggregans* (X99556), *Pyrolobus fumarii* (X99555), *Methanobacterium bryantii* (M59124), *Methanosphaera stadtmanii* (M59139), *Methanobrevibacter ruminantium* (AY196666), *Methanobrevibacter smithii* (AF054208), *Methanobrevibacter gottschalkii* PG (U55238) and *Methanopyrus kandleri* (AE010349).

Results and discussion

A total of 108 clones were examined from the pooled PCR products from four purebred Murrah buffaloes (*Bubalus bubalis*) from India, and 94.42% of the clones (102 of 108 clones) belonged to *Methanomicrobium mobile*. This finding supports our hypothesis that the genus *Methanomicrobium* is the most dominant component of the methanogen populations in Murrah buffaloes (*Bubalus bubalis*). The high percentage obtained from the present study is in contrast to the investigation by Wright *et al.* (2004) in which the genus *Methanobrevibacter* accounted for over 97% of the total number of clones identified from sheep in Western Australia and also different from other studies in which *Methanobrevibacter* strains accounted for the majority of methanogens in the rumen (Miller and Wolin 1986; Sharp *et al.* 1998; Whitford *et al.* 2001; Wright and Pimm 2003; Irbis and Ushida 2004; Skillman *et al.* 2004, 2006; Wright *et al.* 2004, 2006, 2007).

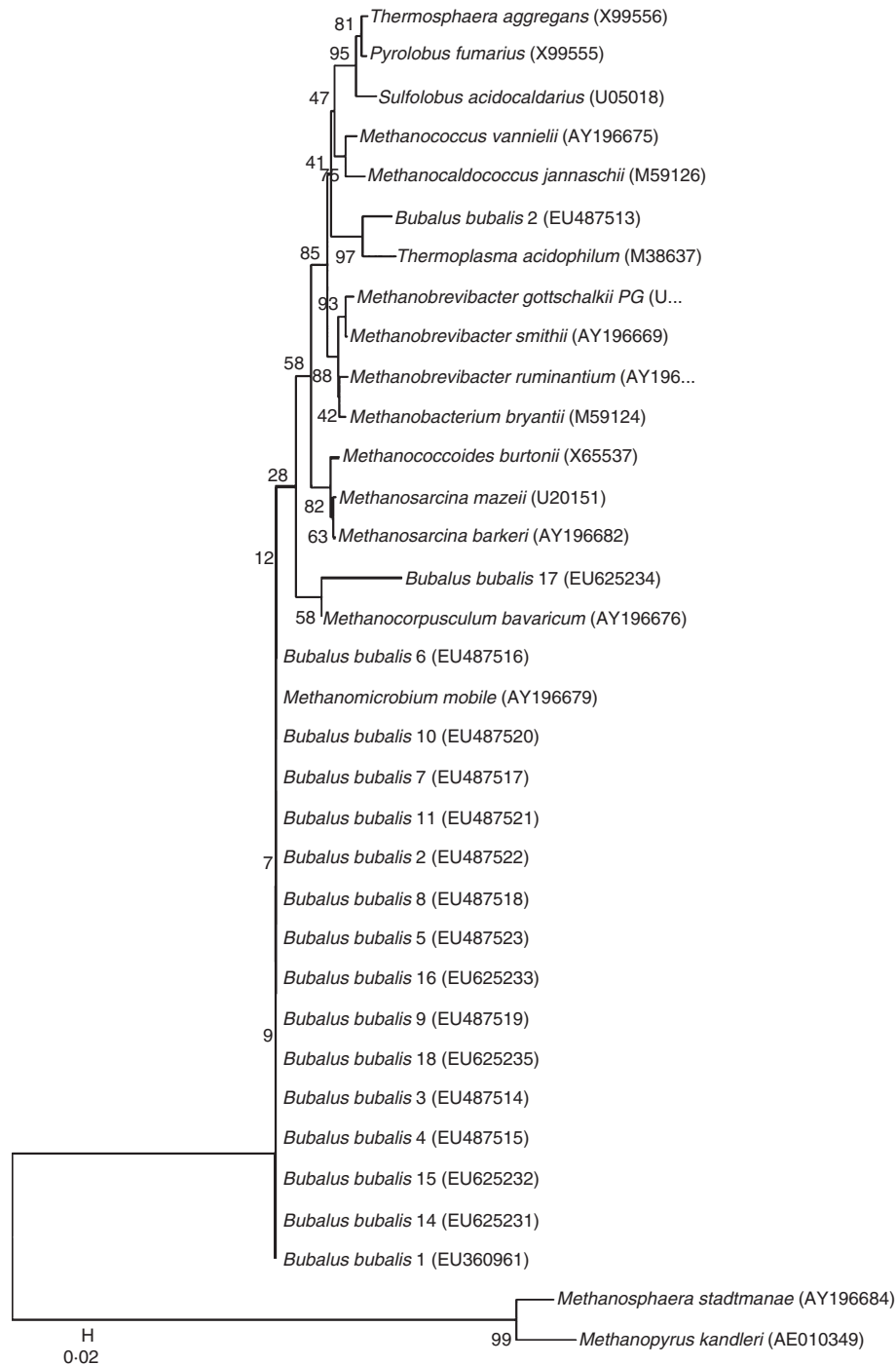


Figure 1 Evolutionary relationships of 34 taxa. The evolutionary history was inferred using the Neighbour-Joining method (Saitou and Nei 1987). The optimal tree with the sum of branch length 4.21980217 is shown. The percentage of replicate trees in which the associated taxa clustered together in the bootstrap test (500 replicates) is shown next to the branches (Felsenstein 1985). The tree is drawn to scale, with branch lengths in the same units as those of the evolutionary distances used to infer the phylogenetic tree. The evolutionary distances were computed using the Maximum Composite Likelihood method and are in the units of the number of base substitutions per site (Tamura *et al.* 2004). Codon positions included were first + second + third + Noncoding. All positions containing gaps and missing data were eliminated from the dataset (complete deletion option). There were 373 positions in the final dataset. Phylogenetic analyses were conducted in mega4 (Tamura *et al.* 2007).

However, in contrast to previous studies, none of the 16S rRNA gene clones generated from the present study branched within the clade consisting of uncultured archaea (Tajima *et al.* 2001; Wright *et al.* 2004, 2006; Snell-Castro *et al.* 2005). The 108 clones examined in the present study revealed 17 different sequences or phylotypes. BLAST and CLUSTAL W alignment analyses of the rumen contents from individual animals also revealed 17 different sequences with a range of 3–10 sequences per animal. Phylogenetic analysis (Fig. 1) revealed that 12 phylotypes (102 clones) were more than 98% similar to *Methanomicrobium mobile*, whereas one clone sequence clustered with *Thermoplasma acidophilum* and one clone sequence clustered with *Methanocorpusculum bavaricum* during the phylogenetic analysis. In closing, the geographical separation of ruminant livestock in India did appear to have an effect on methanogen diversity as none of the phylotype from the present study was 100% similar to the methanogen strain from ruminants in other parts of the world. Data from the present study signify that *Methanomicrobium* phylotype have been found to be the dominant component of the methanogen populations in Murrah buffaloes (*Bubalus bubalis*) from India when compared with the dominance of *Methanobrevibacter* phylotype in Australia. It is likely that *Methanomicrobium* phylotype is the dominant methanogen phylotype in buffaloes worldwide.

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